MAKERERE UNIVERSITY | COLLEGE OF COMPUTING & INFORMATION SCIENCES

SYSTEM ARCHITECTURE & INTERFACE CONTRACT (design.md)

Hardware Pinout, Wireless ESP-NOW Protocol & SD Card Schemas

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Project: Smart Control and Monitoring System for CoCIS

Target Platform: ESP32 (ESP-IDF) + Wokwi Simulation

Communication Bus: Wireless ESP-NOW (2.4 GHz Peer-to-Peer Wi-Fi)

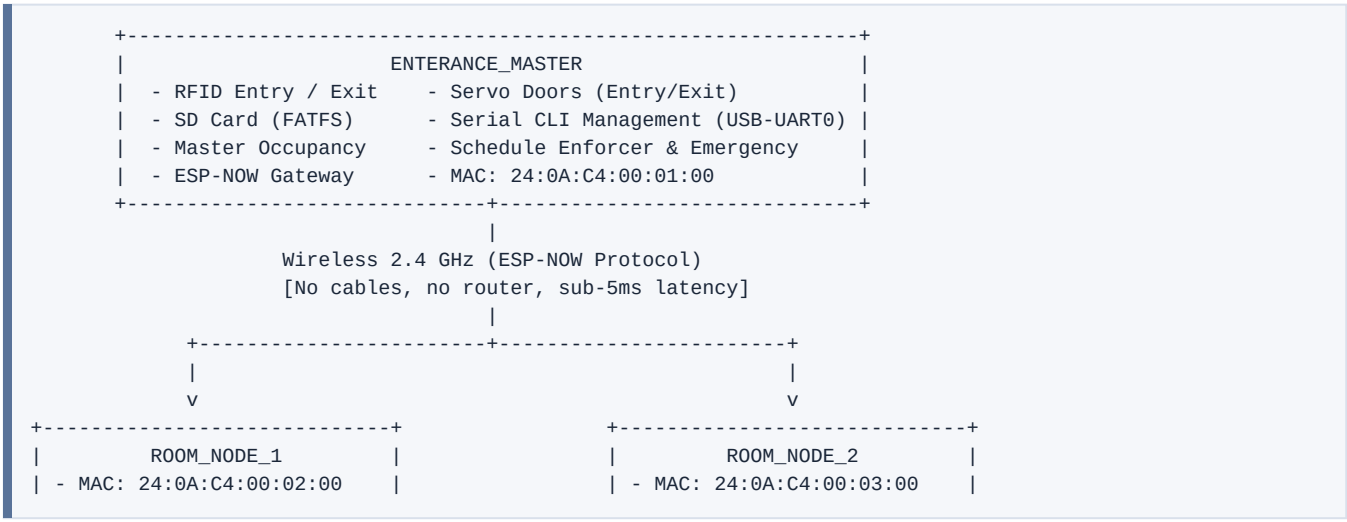
Node Designations: ENTERANCE_MASTER, ROOM_NODE_1, ROOM_NODE_2

Status: Frozen for Team Implementation

1. System Topology Overview (Wireless Star Architecture)

The system consists of three ESP32 microcontroller nodes communicating wirelessly using ESP-NOW (2.4 GHz connectionless Wi-Fi):

- 1. ENTERANCE_MASTER (Entrance Terminal & Central Gateway - ESP32 #1):
- Handles entry & exit RFID scanners, servo door actuators, buzzer, and feedback LEDs.
- Hosts the SD Card module (SPI/FATFS) for logging, user database, and configuration storage.
- Provides the Serial CLI over USB-UART0 for administrative and monitoring access.
- Acts as the central synchronization master and wireless gateway, receiving telemetry from Room nodes and dispatching lighting/emergency overrides via ESP-NOW.
- 2. ROOM_NODE_1 (Lecture Room 1 Controller - ESP32 #2):
- PIR motion detection, ambient light sensing (LDR), and light relay control.
- Transmits room occupancy, lux levels, and lighting status to ENTERANCE_MASTER wirelessly via ESP-NOW.
- 3. ROOM_NODE_2 (Lecture Room 2 Controller - ESP32 #3):
- Identical hardware setup to ROOM_NODE_1; manages Lecture Room 2 and transmits wireless telemetry to ENTERANCE_MASTER.



```
| - PIR Motion Sensor (GPIO14)|                | - PIR Motion Sensor (GPIO14)|
| - Light Relay / LED (GPIO12)|                | - Light Relay / LED (GPIO12)|
| - Ambient LDR (ADC GPIO34)  |                | - Ambient LDR (ADC GPIO34)  |
| - ESP-NOW Wireless Sender   |                | - ESP-NOW Wireless Sender   |
+-----------------------------+                +-----------------------------+
```

2. Hardware Pin Allocations

ENTERANCE_MASTER Pin Mapping (ESP32 #1)

Component / Function	ESP32 GPIO Pin	Description / Interface
RFID Entry Reader	GPIO 21 (SDA), GPIO 18 (SCK), GPIO 23 (MOSI), GPIO 19 (MISO), GPIO 22 (RST)	SPI Bus / RC522 (or Serial Emulation)
RFID Exit Reader	GPIO 5 (SDA), Shared SPI Bus (SCK:18, MOSI:23, MISO:19), GPIO 4 (RST)	SPI Secondary Chip Select
Entry Door Actuator	GPIO 13	PWM / Servo Motor (0 Closed, 90 Open)
Exit Door Actuator	GPIO 15	PWM / Servo Motor (0 Closed, 90 Open)
Access Granted LED	GPIO 2	Green LED (Active High)
Access Denied LED	GPIO 4	Red LED (Active High)
Alarm Buzzer	GPIO 12	Active Buzzer (Lockout & Alarm)
SD Card CS	GPIO 27	SPI CS (Shared VSPI or Dedicated HSPI)
SD Card MOSI	GPIO 23	SPI MOSI
SD Card MISO	GPIO 19	SPI MISO
SD Card CLK	GPIO 18	SPI SCK
Serial CLI Terminal	TX0 (GPIO 1), RX0 (GPIO 3)	115200 Baud (USB-Serial Monitor)
Wireless Transceiver	Internal RF Antenna	2.4 GHz ESP-NOW Wireless Mesh/Star
Emergency Pushbutton	GPIO 32	Pull-up Input (Active Low)

ROOM_NODE_1 & ROOM_NODE_2 Pin Mapping (ESP32 #2 & #3)

Component / Function	ESP32 GPIO Pin	Description
PIR Motion Sensor	GPIO 14	Digital Input (Interrupt driven)
Light Relay / Lamp LED	GPIO 12	Digital Output (Active High)
LDR Light Sensor	GPIO 34	Analog Input (ADC1_CH6)
Wireless Transceiver	Internal RF Antenna	2.4 GHz ESP-NOW Telemetry Link
Status LED	GPIO 2	Blinks on wireless packet transmission

3. Wireless ESP-NOW Protocol & Packet Framing (‘common/protocol.h‘)

ESP-NOW transmits variable-length payloads up to 250 bytes encapsulated within 802.11 Vendor-Specific Action Frames. All application packets follow this standardized structure:

```
+--------------+--------+-----------+-----------+----------+-------------+---------+----------
| MAGIC (2B)    | MSG_ID | SENDER_ID | TARGET_ID | MSG_TYPE | PAYLOAD_LEN | PAYLOAD | CHECKSUM
| 0xAA 0x55     | 1 byte |  1 byte   |  1 byte   |  1 byte  |   1 byte    | N bytes |  1 byte
+--------------+--------+-----------+-----------+----------+-------------+---------+----------
```

Peer Device Addressing

- ENTERANCE_MASTER: Target ID 0x01 | Default MAC: 24:0A:C4:00:01:00
- ROOM_NODE_1: Target ID 0x02 | Default MAC: 24:0A:C4:00:02:00
- ROOM_NODE_2: Target ID 0x03 | Default MAC: 24:0A:C4:00:03:00
- Broadcast Address: FF:FF:FF:FF:FF:FF (Used for emergency evacuation alerts)

Message Identifiers:

- 0x01 : MSG_TYPE_HEARTBEAT (Periodic keep-alive ping sent every 2 seconds)
- 0x02 : MSG_TYPE_ROOM_STATUS (Telemetry: uint8_t occupied, uint8_t lights_on, uint16_t lux, uint32_t active_sec)
- 0x03 : MSG_TYPE_LIGHT_OVERRIDE (Master command: 0=Auto, 1=Force OFF, 2=Force ON)
- 0x04 : MSG_TYPE_EMERGENCY_ALERT (Broadcast command: 1=Evacuate, 0=Normal)

Delivery Verification & Reliability

- 1. Hardware MAC Acknowledgement: Every unicast ESP-NOW frame triggers a hardware ACK within the 802.11 MAC layer. The sender's esp_now_register_send_cb callback confirms delivery status (ESP_NOW_SEND_SUCCESS vs ESP_NOW_SEND_FAIL).
- 2. Automatic Retries: If ESP_NOW_SEND_FAIL is detected, the sender retries up to 3 times before raising a communication fault.
- 3. Offline Detection: If ENTERANCE_MASTER receives no heartbeat or telemetry from a room node for more than 10 seconds, it flags that room as OFFLINE on the serial console.

4. Storage Schemas & File Formats (SD Card)

4.1 '/sdcard/users.csv'

Contains the authorized registry of users:

```
tag_id,name,role,is_active
TAG_1001,Kayeera Nathan,admin,1
TAG_1002,Kasozi David,staff,1
TAG_1003,Zziwa Mark,student,1
TAG_1004,Sseku Jovan,student,1
TAG_1005,Namuzibwa Laurinda,staff,1
```

4.2 '/sdcard/access_log.csv'

Appended chronologically on every swipe event:

```
timestamp,tag_id,user_name,direction,decision,reason
2026-10-24 08:15:20,TAG_1001,Kayeera Nathan,ENTRY,GRANTED,Authorized
2026-10-24 21:05:11,TAG_1003,Zziwa Mark,ENTRY,DENIED,Outside Schedule
2026-10-24 21:10:02,TAG_1004,Sseku Jovan,EXIT,DENIED,Curfew Lockdown
2026-10-24 21:15:00,SYSTEM,Admin,EMERGENCY,ACTIVE,Evacuation Triggered
```

4.3 '/sdcard/occupancy.csv' (For Crash Recovery)

Updated dynamically so building occupancy survives unexpected reboots:

```
tag_id,user_name,entry_timestamp
TAG_1001,Kayeera Nathan,2026-10-24 08:15:20
```

4.4 ‘/sdcard/config.txt‘

```
schedule_start=07:00
schedule_end=20:00
lockout_attempts=3
lockout_duration_sec=60
no_motion_timeout_sec=30
heartbeat_interval_sec=2
```

5. Serial CLI Command Specification

Command	Arguments	Permissions	Output / Action
help	none	Public	Displays list of available commands
status	none	Public	Shows system health, wireless room links, SD status
occupancy	none	Public	Lists all persons currently recorded inside
rooms	none	Public	Reports live wireless telemetry from ROOM_NODE_1 & 2
lights	none	Public	Detailed illumination, lux levels & relay states
login	<user> <pass>	Public	Authenticates admin session
logout	none	Authenticated	Terminates active session
schedule show	none	Public	Shows current entry/exit permitted hours
schedule set	<HH:MM> <HH:MM>	Admin Only	Updates building operating hours in RAM & SD
logs	[count]	Admin Only	Dumps the last N log entries
adduser	<tag> <name> <role>	Admin Only	Appends new card to users.csv
deluser	<tag>	Admin Only	Deactivates card in users.csv
energy	none	Admin Only	Displays computed energy savings (kWh)
emergency	‘<on\	off>‘	Admin Only

6. Access Control & Schedule Rules Engine

The decision logic executed on ENTERANCE_MASTER evaluates every tag swipe in the following order:

```
[Tag Scanned]
      |
      v
Is terminal currently locked out (intrusion)?  --YES--> [DENY & Sound Alarm]
      | NO
```

```
Is Tag in users.csv & active? ---------------- NO --> [DENY & Increment Invalid Counter]
      | YES (Reset Invalid Counter)
Is current time within Building Schedule?
      |-- NO --> Is User Role == 'admin'?
      |              |-- NO --> [DENY: Outside Schedule Hours (Applies to ENTRY & EXIT)]
      |              |-- YES -> Bypass Schedule Check
      |-- YES -> Proceed to Anti-Passback Check
                     |
Is direction == ENTRY and user already INSIDE? --YES--> [DENY: Anti-Passback Violation]
Is direction == EXIT and user already OUTSIDE? -YES--> [DENY: Anti-Passback Violation]
                     | NO
             [GRANT ACCESS]
       - Trigger appropriate door servo (unlock for 5s)
       - Blink Green LED / Beep Buzzer
       - Update Occupancy Table (Mutex protected)
       - Push event to SD Log Queue
```